

Tyler Bibus



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Professional Summary

Motivated Computer Engineering undergraduate at Iowa State University (GPA: 3.9) with hands-on experience in 5G networking, Android development, and processor design. Skilled in C++, VHDL, and Java, with a proven track record in hand-on projects. Eager to contribute to innovative tech projects through internships or entry-level roles.

Career Experience

CPRE 3810 Teaching Assistant – Iowa State University, Ames IA August 2025 – present

- Overseeing 46 students in lab for computer architecture, assisting students with any issues they have related to computer architecture.
- Assisting professor with conversion from MIPS based architecture to RISC-V, including documentation as well as our testing framework for students.

Undergraduate Researcher – ARA Wireless, Ames IA January 2025 – August 2025

- Collaborating directly under a graduate mentor implementing 5G networking outdoors, with a focus on bringing high speed and reliable internet to rural areas.
- Updated existing 5G software to work outdoors with amplifiers, utilizing both C and C++ to utilize output pins of a software-defined radio (SDR) resulting in a 48dB (10⁴⁸X) signal improvement.
- Tested and implemented Mender, a software update scheduling software that allows for remote updates on our towers across Ames, rather than manually scaling towers to retrieve and update systems.

Undergraduate Research Programmer - CSAFE, Ames IA October 2023 – May 2024

- Collaborated with a diverse team of remote researchers to develop a tool for Android digital forensics (EviHunter).
- Developed a web crawler to scrape several hundred Android APK files and metadata for the EviHunter project database.

CAD Technician – Americom, Eagan MN August 2021-August 2023

- Designed and drafted well over 150 AV, network cabling, and security drawings for technicians using AutoCAD LT.
 - Collaborated with sales and design teams to ensure accurate and quality drawings that met install technician and client standards.
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Education and Awards

Bachelor's degree in Computer Engineering with Minor in Music, Iowa State University, Ames, IA

Expected Graduation: December 2026 | GPA: 3.9

- President's Award (Perfect 4.0 for 2 consecutive semesters starting fall 2024).
 - Dean's List Recipient - 4 Time
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Projects

Goods and Service Finder (Android App), Spring 2025

- Created an Android app using RESTful APIs with a backend using spring boot.
- Implemented concrete UI/UX design using Android XML and fragment layouts to create a fluid and performant application.

MIPS Processor (VHDL Implementation), Spring 2025

- Designed a single-cycle processor adhering to MIPS ISA with 100% instruction coverage.

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- Developed software-scheduled and hardware-scheduled pipelined versions, achieving significant performance gains over single-cycle design.

Personal Websites (Websites using React and TailwindCSS), Spring 2025

- Website uses a hash router system that streamlines navigation with GitHub Pages.
- Utilizes good UI/UX standards with as much cross platform compatibility as possible.

Local Discord LLM Bot, Summer 2025

- Personal passion project utilizing LM Studio's APIs in python to create a customizable Discord bot.
- LLM Ran locally on my computer with personal optimizations to ensure timely responses.

AI Hardware Accelerator, Fall 2025

- Part of my CPRE 5870 – Hardware Design for Machine Learning course, I had to create a Hardware accelerator using a FPGA board for an image classification model.
- Compared several different fast multipliers with different quantization sizes, demonstrating the findings to the class while tying our project to related research papers.

Other Activities

- **Food At First** – Weekly volunteering at a local food shelf to disperse food to the community.
- **Symphonic Orchestra** – Part of Iowa State University's symphonic orchestra as a Bass player.
- **ChipForge** – Part of the ChipForge club at Iowa State University, an ASIC design club where I learn more about industry standards for ASIC design, testing, and fabrication.

Skills

- **Tools** – AutoCAD LT, Android Studio, Unity, QuestaSim, Sphinx, CI/CD, Docker, Kubernetes, ModelSim, LT Spice, TensorFlow, ASIC, FPGA Boards, AI, AI Model Training, AI Model Quantization
- **Programming Languages** – C, C++, Java, JavaScript, HTML, Python, Verilog, VHDL, MIPS Assembly